Data Flow Diagram

Date	7 July 2026
Team ID	**SWUID2026221619**
Project Name	**A Comprehensive Measure of Well-Being**
Maximum Marks	2 Marks

Data Flow Diagram

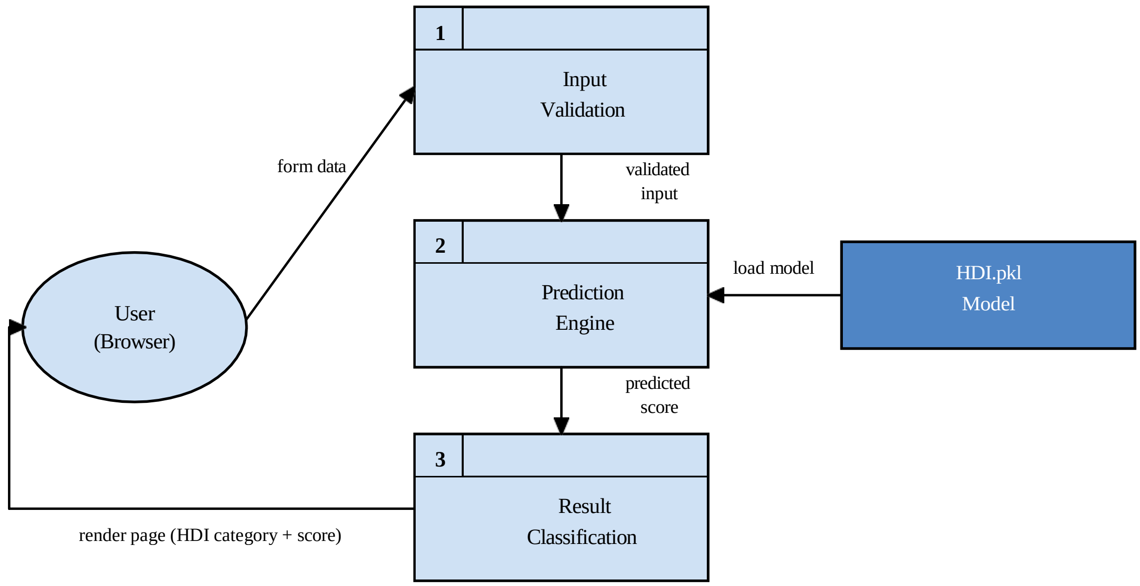
Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol Name	Description
External Entity (Oval)	A person, organization, or system outside the project that sends or receives data (e.g., User, Browser).
Process (Numbered Rectangle)	An activity that transforms incoming data into outgoing data (e.g., Validate Input, Predict HDI).
Data Store (Solid Rectangle)	A place where data is held for later use (e.g., HDI.csv Dataset, HDI.pkl Model).
Data Flow (Labeled Arrow)	The movement of data between entities, processes, and data stores, labeled with what data is passed.

HDI Predictor — Data Flow Diagram

The diagram below shows how a user's input flows through validation, the prediction engine, and back as a classified result.



Level 1 — Detailed Process Table

Process	Input	Processing Steps	Output	Data Store
1. Data Loading	HDI.csv file	pd.read_csv('HDI.csv') Explore shape, head()	Development DataFrame	dataset/HDI.csv
2. EDA	Development DataFrame	Strip plots, Heatmap Correlation matrix	Visualizations (PNG)	plots/
3. Preprocessing	Development DataFrame	Select features iloc[:,] fillna(mean) LabelEncoder	Cleaned X, y	dataset/X_clean.csv
4. Train/Test Split	Cleaned X, y	train_test_split (test_size=0.1)	x_train, x_test, y_train, y_test	Memory
5. Model Training	x_train, y_train	LinearRegression .fit(x_train, y_train)	Trained reg object	Memory
6. Evaluation	x_test, y_test	reg.predict(x_test) r2_score → 93.74%	R-squared Score, y_pred	plots/actual_vs_predicted.png
7. Model Save	Trained reg object	pickle.dump(reg, open('HDI.pkl','wb'))	HDI.pkl	model/HDI.pkl
8. Flask Predict	User form inputs	model.predict(df) Classify category	prediction_text	resultnew.html